Optimization of Prussian Blue (PB) Analogue-Based Redox-Targeting Flow Batteries Enabled by Automated Redox-Targeting Measurements Using Interdigitated Electrode Arrays (IDAs)

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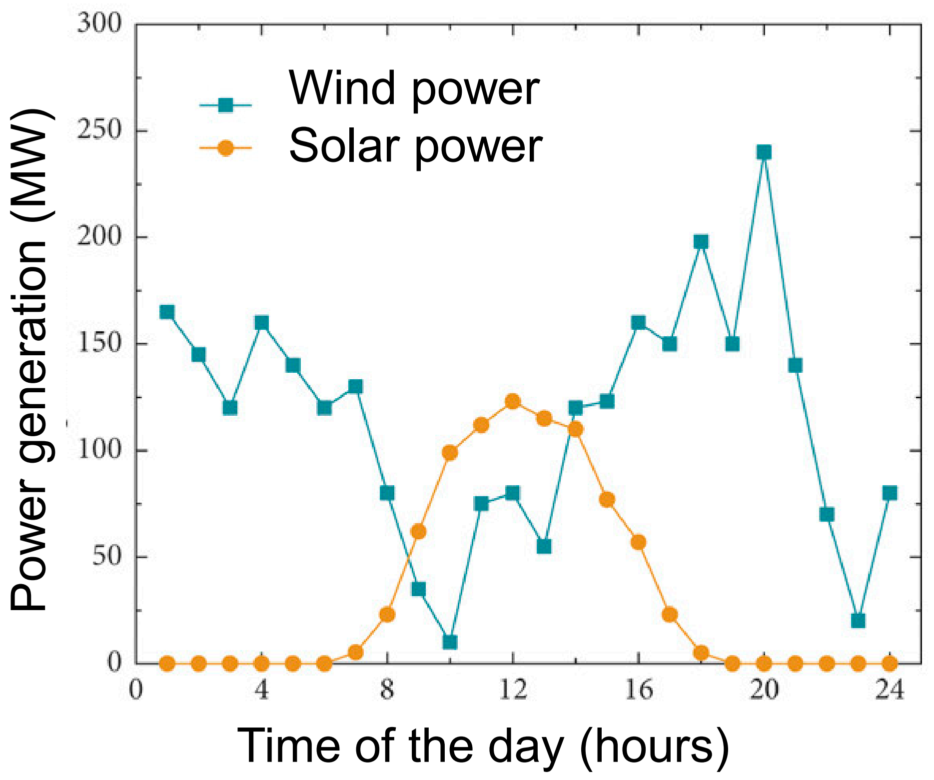
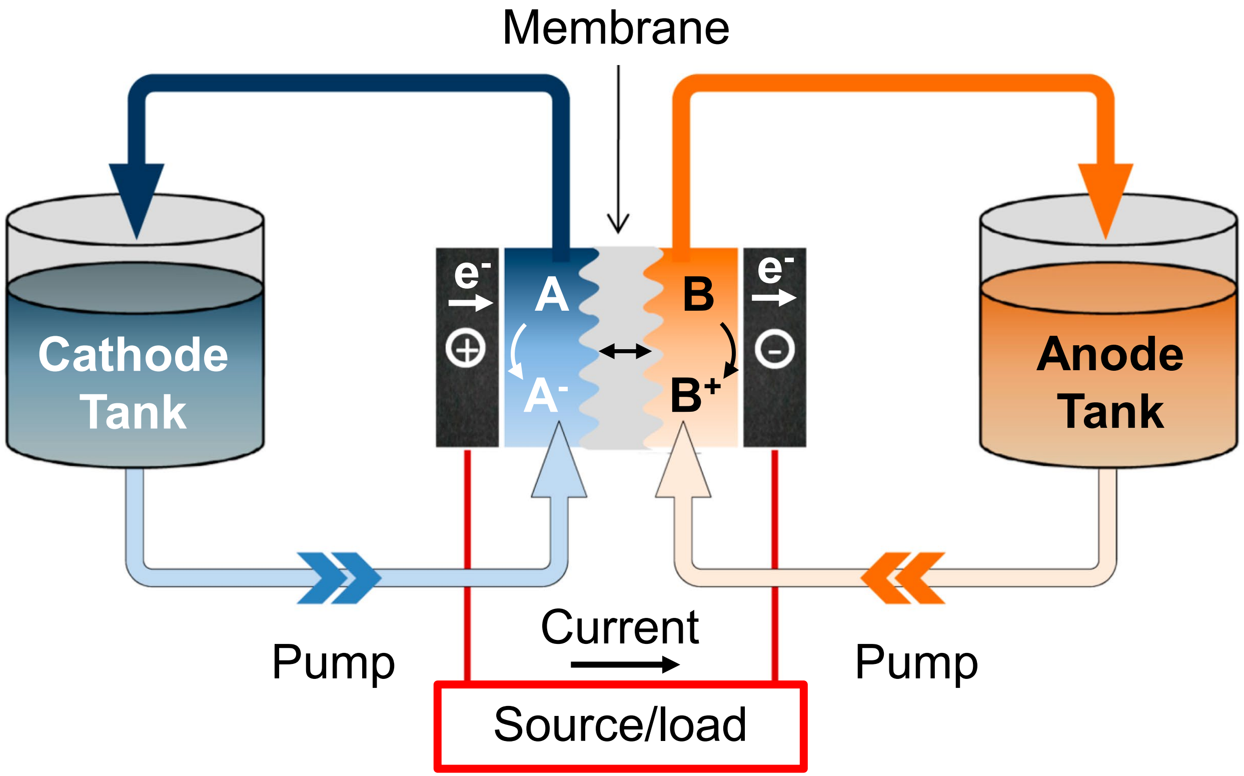
Redox Flow Batteries for Grid-Level Energy Storage



Redox flow battery (RFB)

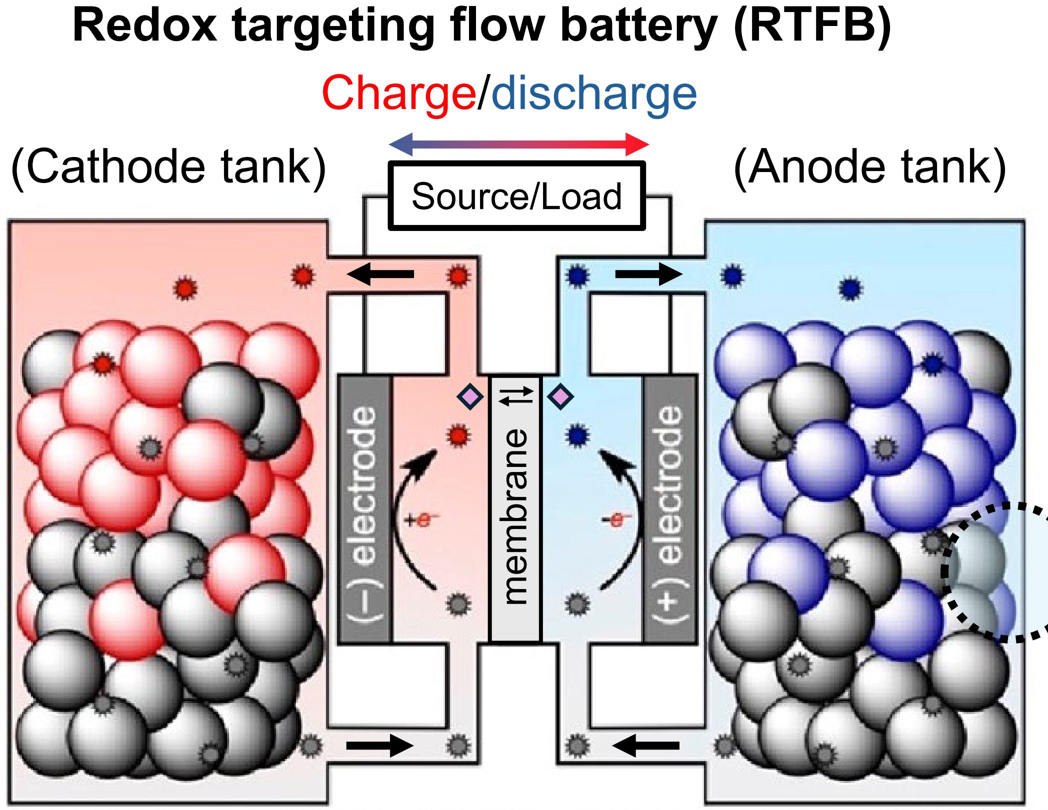


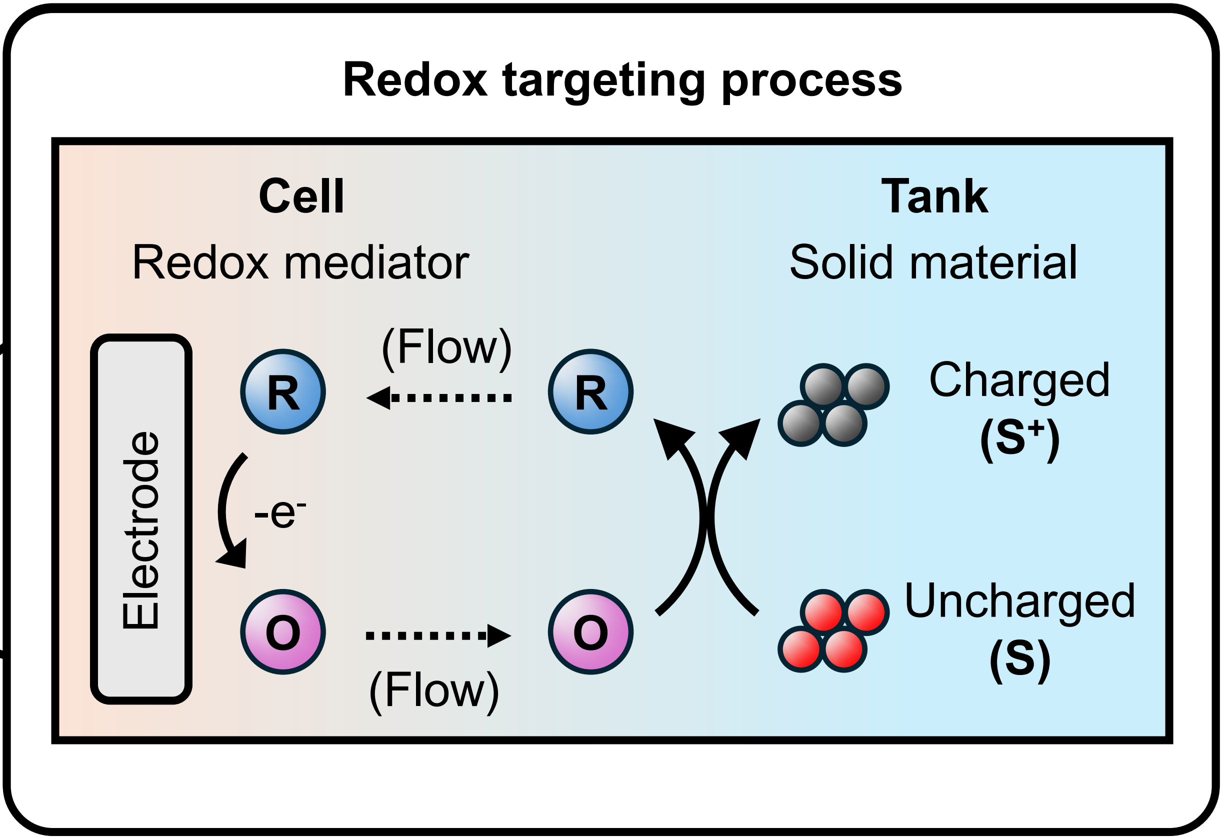
- Renewable energy sources are intermittent

- Highly scalable
- **Limitation:** Energy density is limited by the solubility of redox-active species

Math. Probl. Eng. **2022**, 2022, 1–13
Molecules **2015**, 20 (11), 20499–20517

Redox Targeting Flow Batteries (RTFBs)





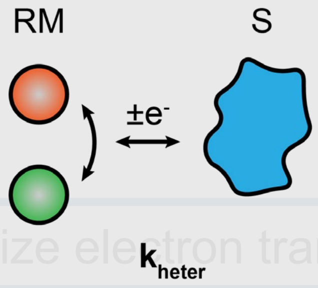
- RTFBs stores energy in a solid material
  - Can incorporate redox-active species up to **20 M**
- Redox mediators (RMs) shuttles charges between the solid and electrode via **redox targeting**

How to study the redox targeting processes?
What are the parameters to optimize?

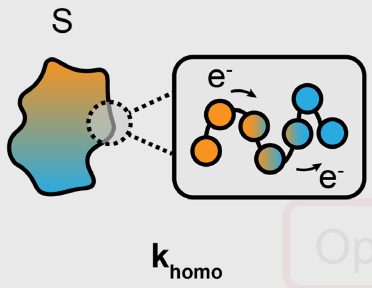
Sci. Adv. **2015**, 1, 10, e1500886
Joule **2019**, *3* (9), 2255–2267

Challenges to Study the Redox Targeting Process

- Heterogenous electron transfer



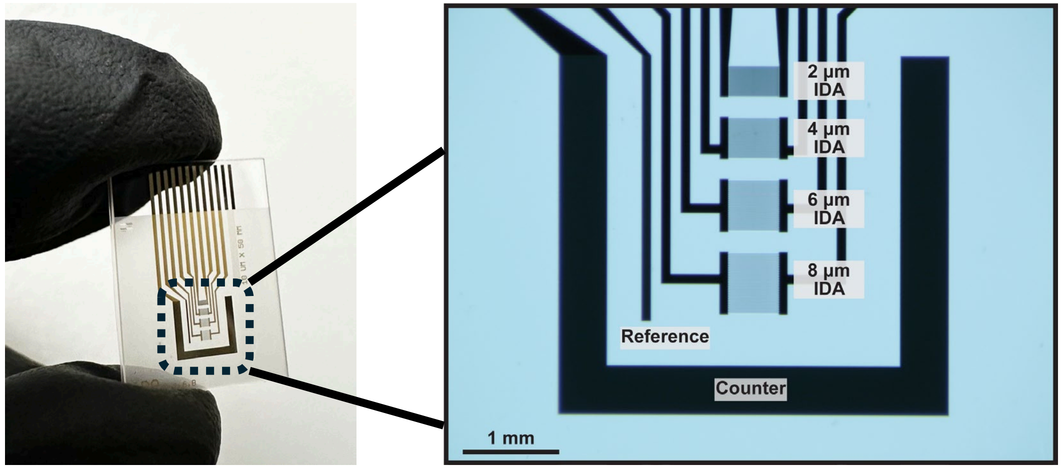
- Homogenous electron transfer



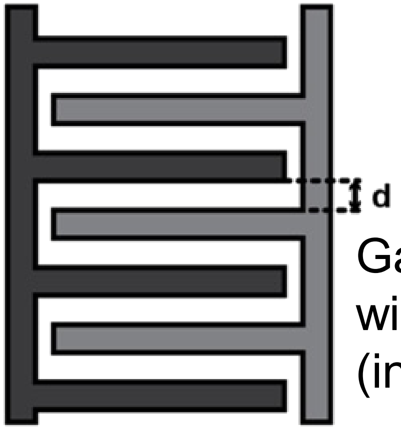
Are we able to develop a platform that offers possibilities to study the **electron transfer process** in RTFBs with a higher throughput?

Chemistry A European J **2022**, 28, 25, e202200149 *Joule* **2019**, *3* (9), 2255–2267

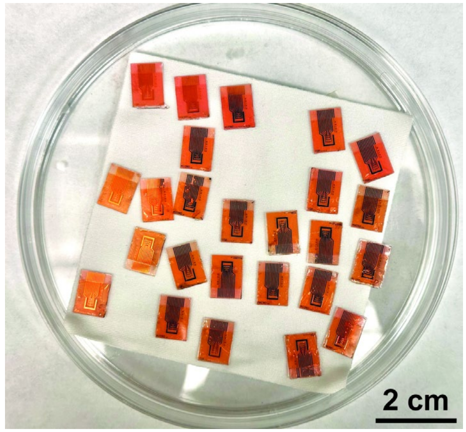
Interdigitated Electrode Arrays (IDAs)



4 IDAs per chip

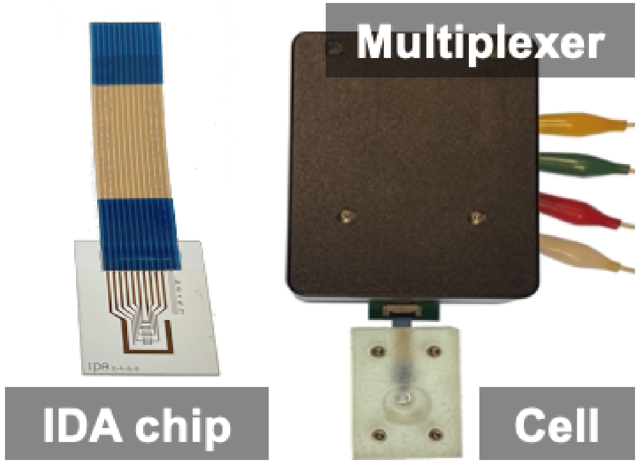


Two **individually addressable** working electrode arrays

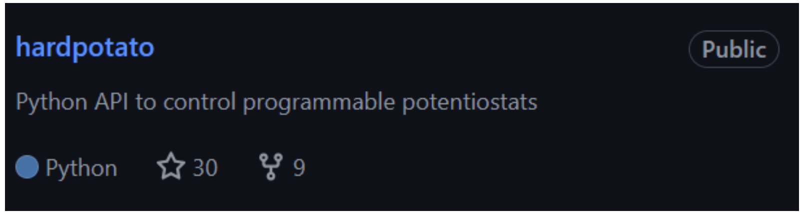


Batched, high-yield microfabrication

Multiplexing circuit to control IDAs



Python library to control multiplexer and commercial potentiostats

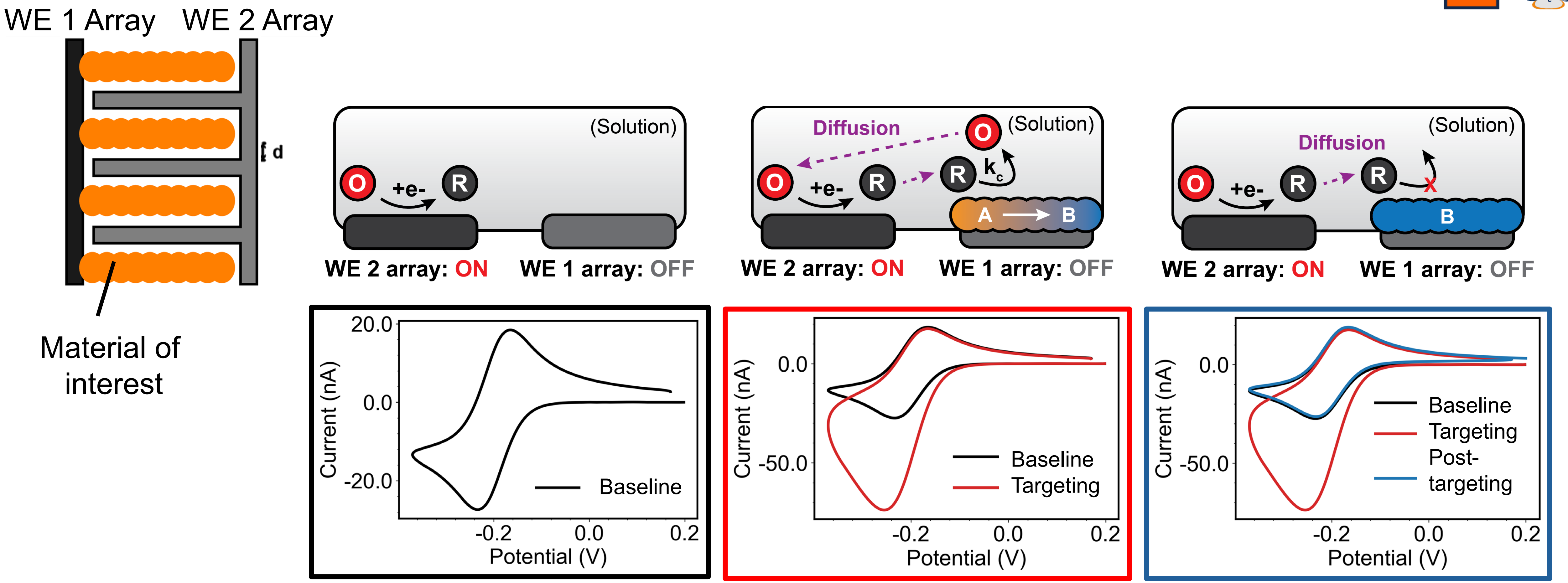


https://github.com/jrlLAB/hardpotato
ACS Meas. Sci. Au **2023**, 3, 1, 62–72
Anal. Chem. **2023**, 95, 11, 4840–4845
Device **2023**, 1, 5, 100103–100103

- **Automated** electrochemical measurements can be performed on IDAs using Python

Check out our GitHub pages!

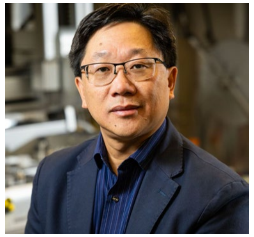
Automated Redox Targeting (ART) with IDAs



- Performing a cyclic voltammetry (CV) with array 2 alone generates a normal “duck-shape” CV
- Pattern the material of interest on array 1
- A **transient current increase** will be observed on the redox targeting CV
- Current level returns to the baseline after the active substrate material has been fully consumed

Previous Work: ART for Multi-Electron Redox Polymers

In collaboration with Pacific Northwest National Laboratory



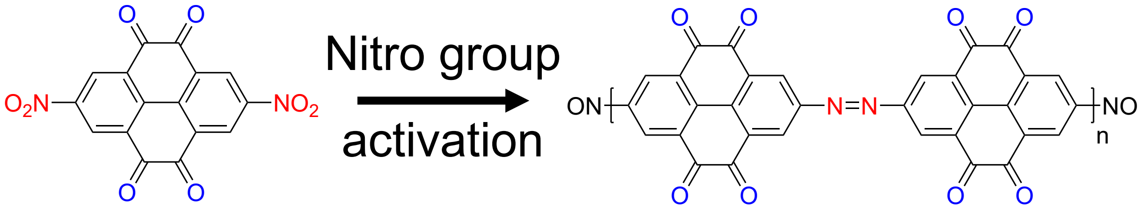
Dr. Wei Wang

Dr. Ruozhu Feng

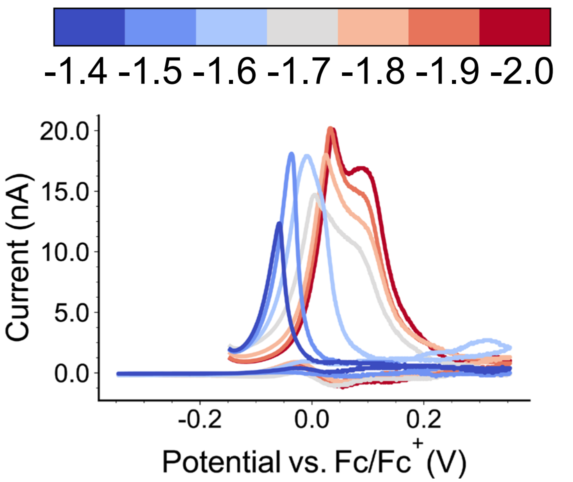
Dr. Eliot Woods

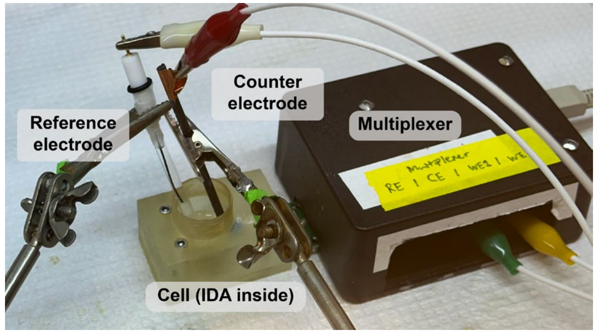
Dinitro pyrene-4,5,9,10-tetrone (PTO)

PTO azo group-based polymer (**PTAP**)

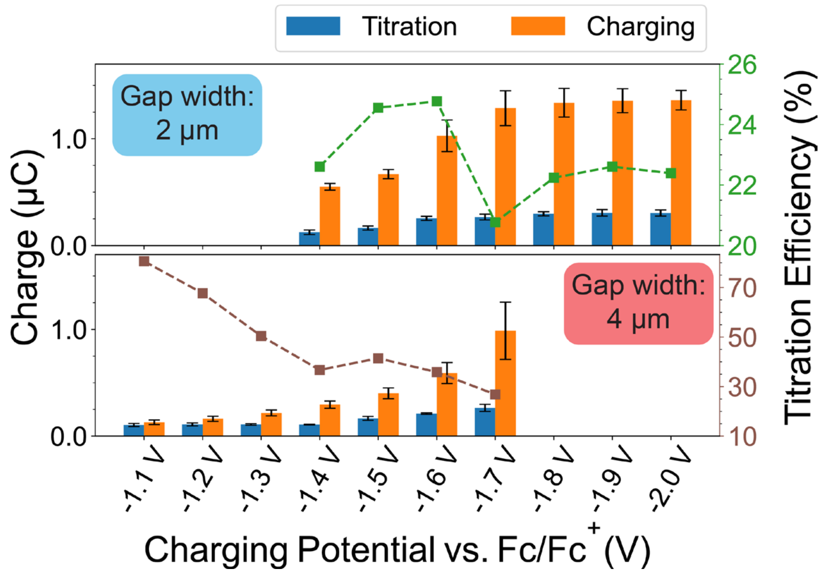


ART inside glovebox





Charge access vs. charging potential



Summary

- Charging potential-dependent ART measurements reveal both **multi-electron** nature of PTAP and significant **chemical irreversibility** of azo group
- Results from ART measurement offer insights for molecular design of multi-electron RAPs

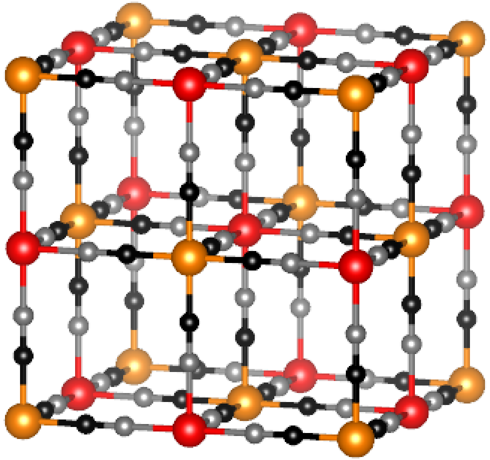
Wang, Z., Pence, M.A., Woods, E.F., Putnam, S.T., Feng, R., Wang, W., Rodríguez-López, J. *ACS Meas. Sci. Au*, ***accepted***

ChemSusChem **2019**, 13, 9, 2449–2456

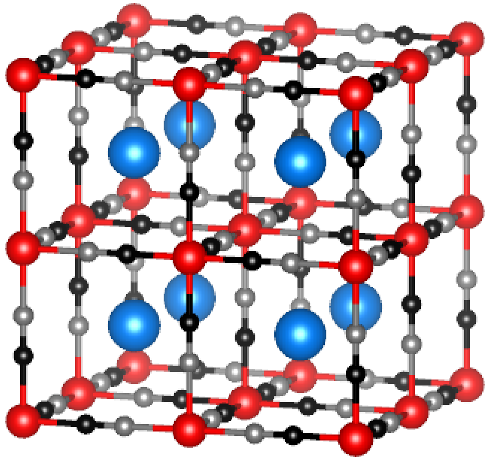
Prussian Blue: Redox and Electrodeposition

Fe^{3+} Fe^{2+} C N Cation

Prussian Blue (PB)
$Fe_4^{3+}[Fe(CN)_6]_3$

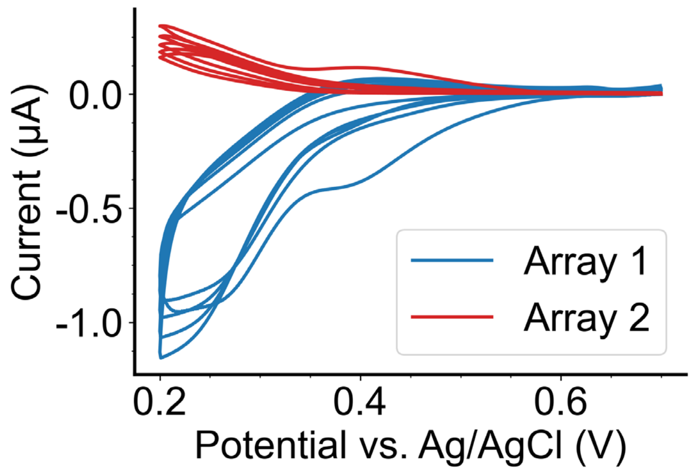


Prussian White (PW)
$Fe_4^{2+}[Fe(CN)_6]_3$

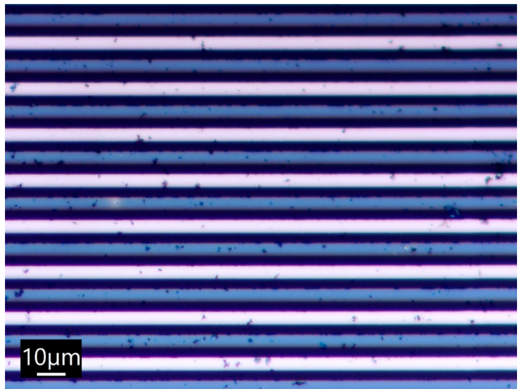


Dual-mode electrodeposition of PB

$$3Fe(CN)_6^{3-} + 3e^- + 4Fe^{3+} \Rightarrow \boldsymbol{Fe_4[Fe(CN)_6]_4}$$



Hypothesis
Electrolyte composition such as cations and ionic strength affects the behavior of PB, which can be revealed by ART approaches in a high-throughput manner.



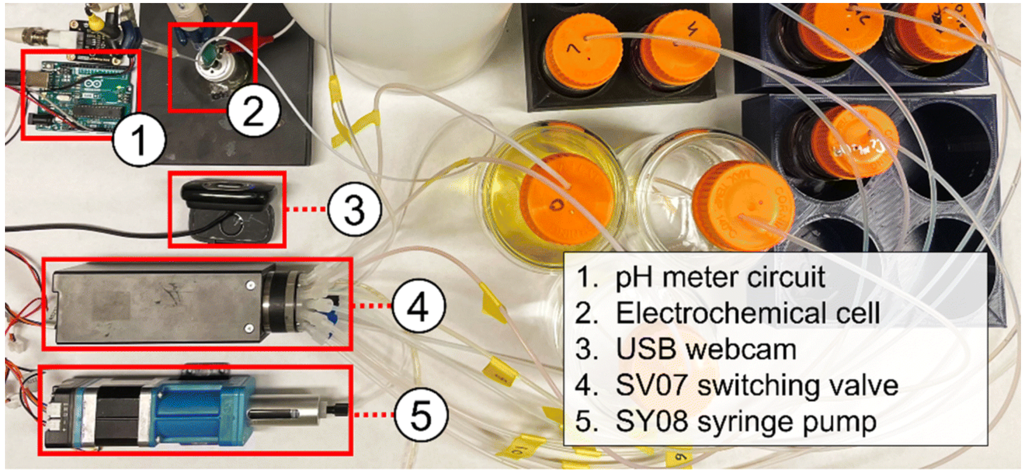
PB thin film (Array 1)

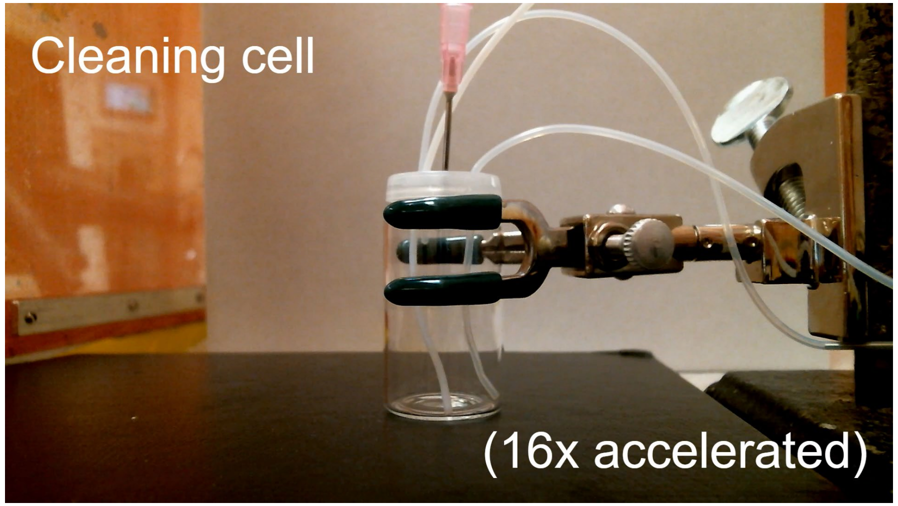
Bare Pt (Array 2)

Joule **2019**, 3, 9, 2255–2267
ACS Energy Lett. **2017**, 2, 3, 615–621
ACS Energy Lett. **2019**, 4, 12, 3028–3035

ART with Solution Handling and Gas Flow Control

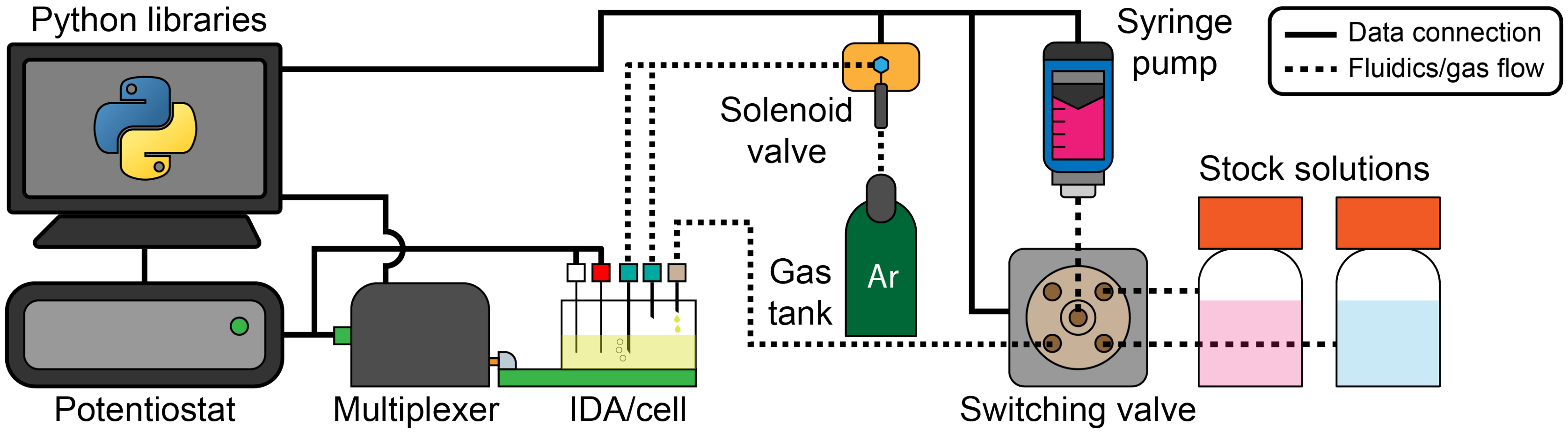
The elab (syringe pump-based solution handling robot) + solenoid valve (Argon flow control)

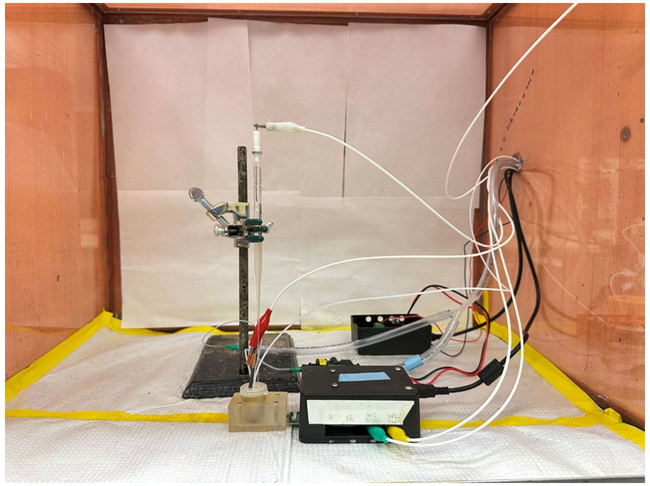




Dr. Michael A. Pence

Gavin Hazen

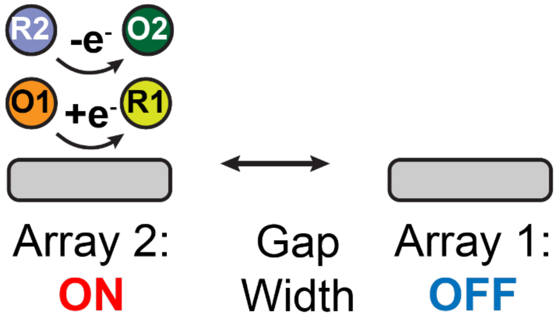


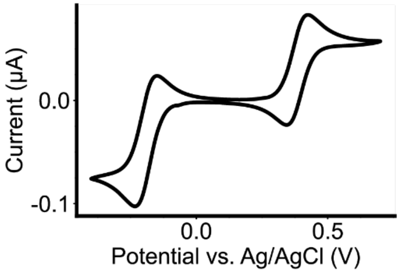


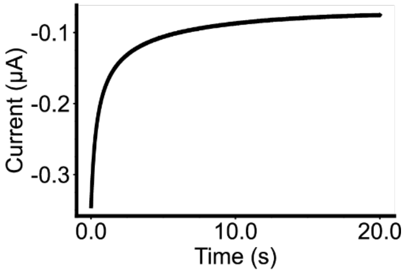
ART of PB: Automated Workflow

1. Baseline Measurement

(Argon bubbling for degassing and resetting the initial condition)

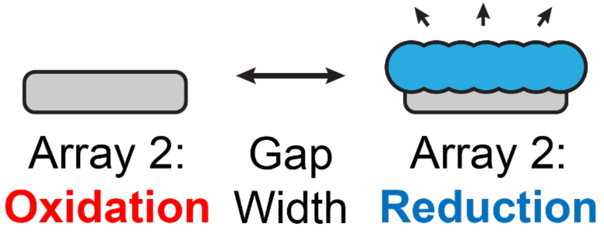


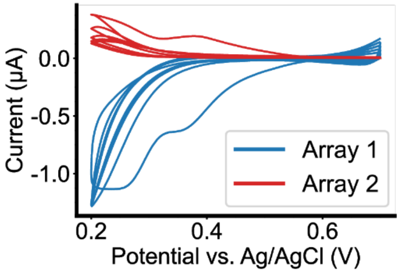




2. PB Electrodeposition

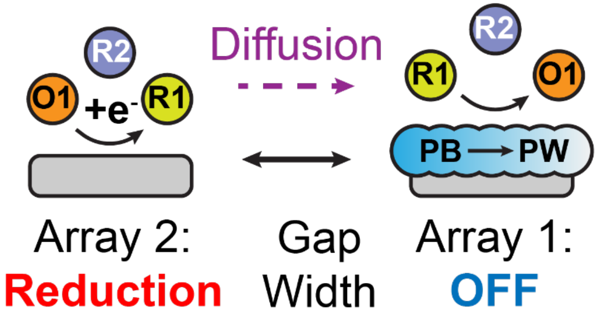
(Precursor preparation and cell cleaning)

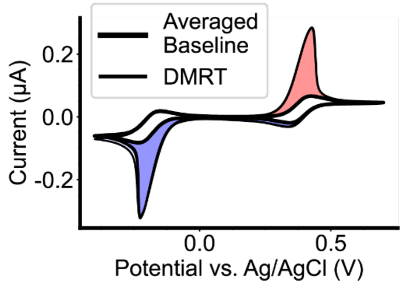


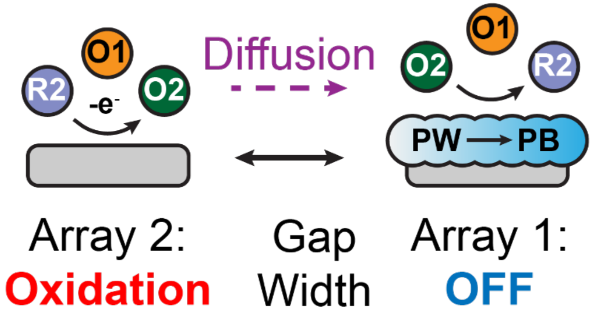


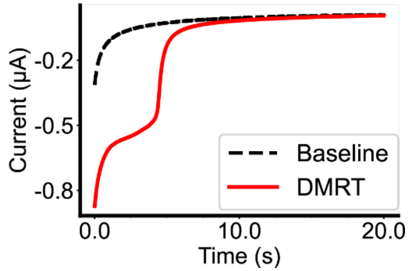
3. Redox Targeting

(Argon bubbling for degassing and resetting the initial condition)



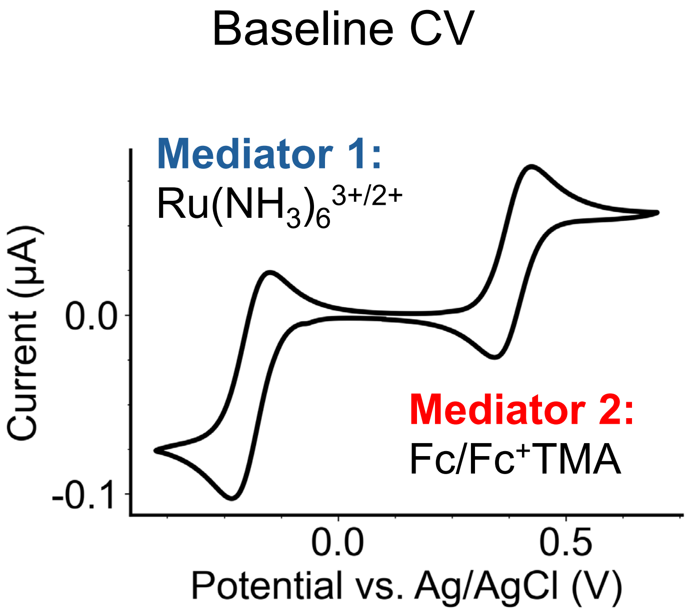


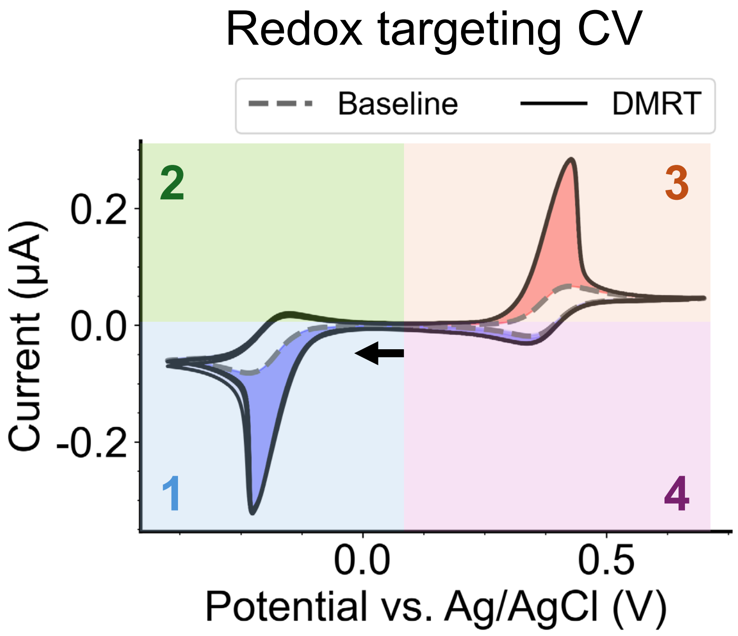


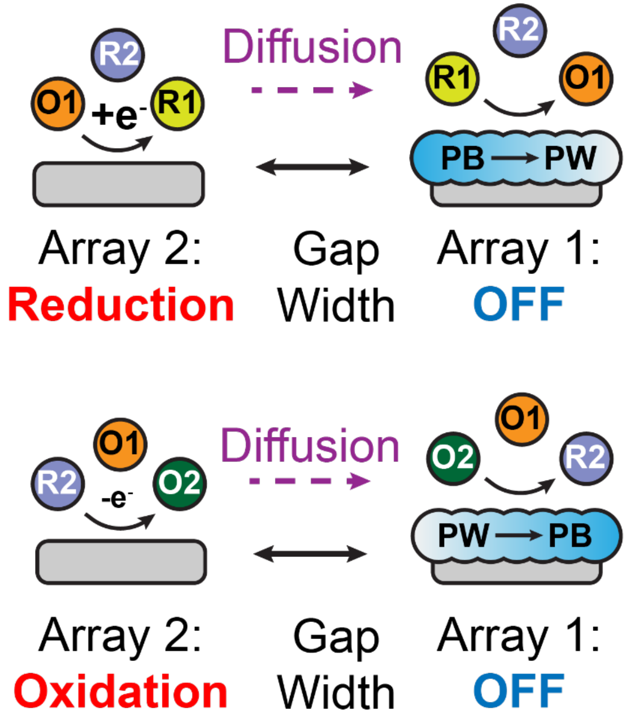


PB's behavior was evaluated in 4 electrolytes ($CsNO_3$, KNO_3, $NaNO_3$, $LiNO_3$) with a dual-mediator redox targeting (DMRT) system.

Redox Targeting CVs



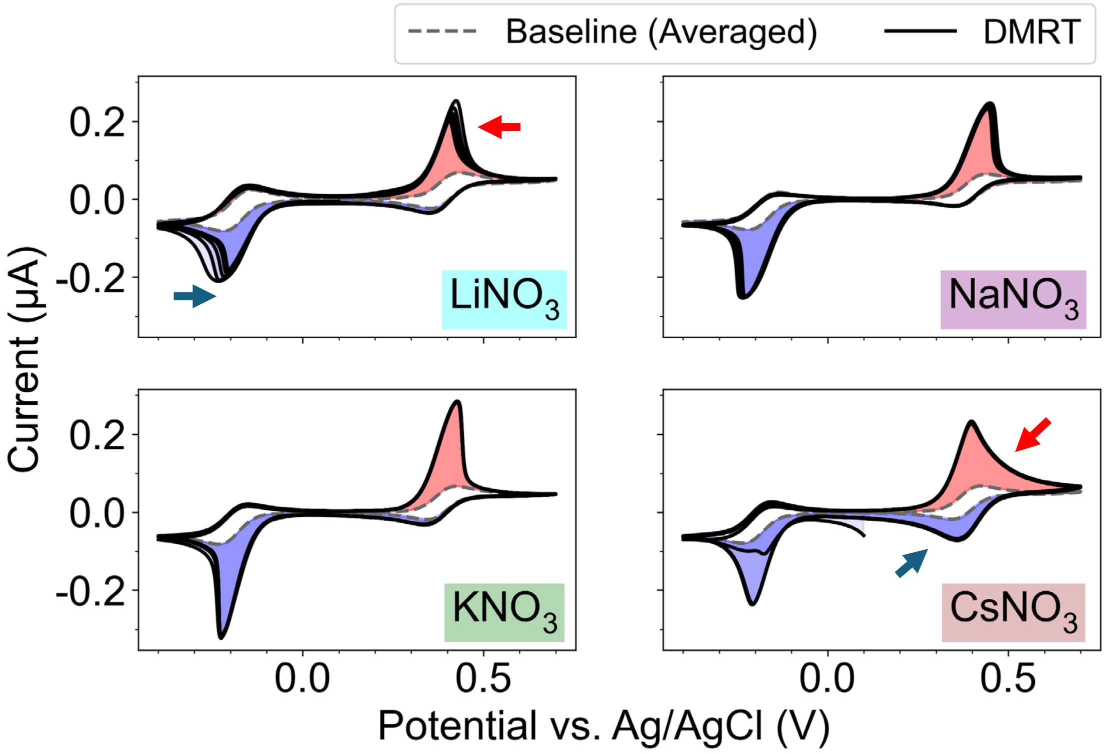


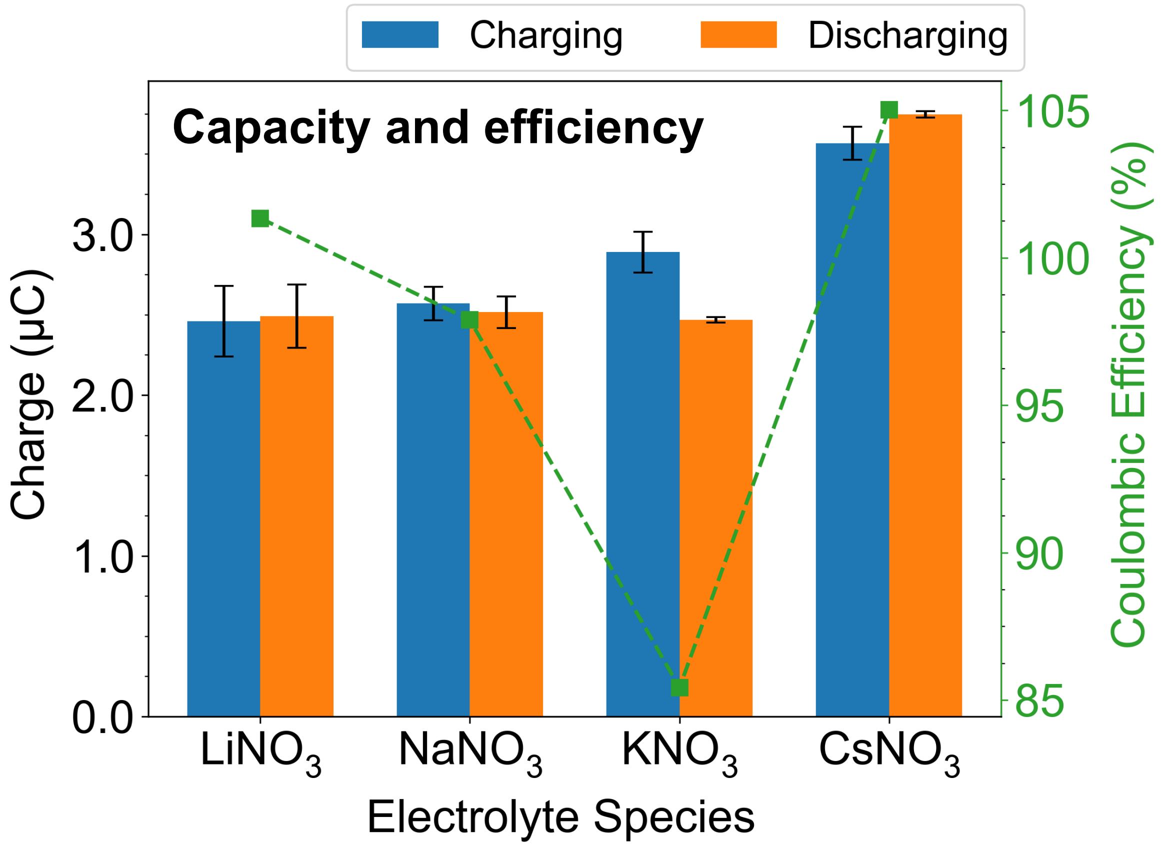


- Initial direction of the CV scan **must be cathodic** because PB is the original state
- 1: Mediator **M1 gets reduced** and **reduces PB to PW** with cation intercalation
- 2: Mediator M1 gets oxidized back to its original form, but is not oxidative enough to convert PW back to PB
- 3: Mediator **M2 gets oxidized** and **oxidizes PW back to PB** with cation deintercalation
- 4: Mediator **M2 gets reduced** back to its original form, but is not reducing enough to convert PB to PW

ART CVs Reveals Electrolyte Effect

ART CVs for all electrolytes



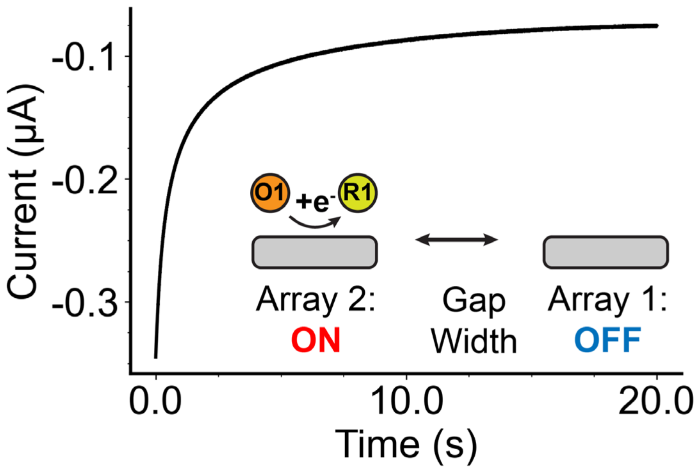


Summary

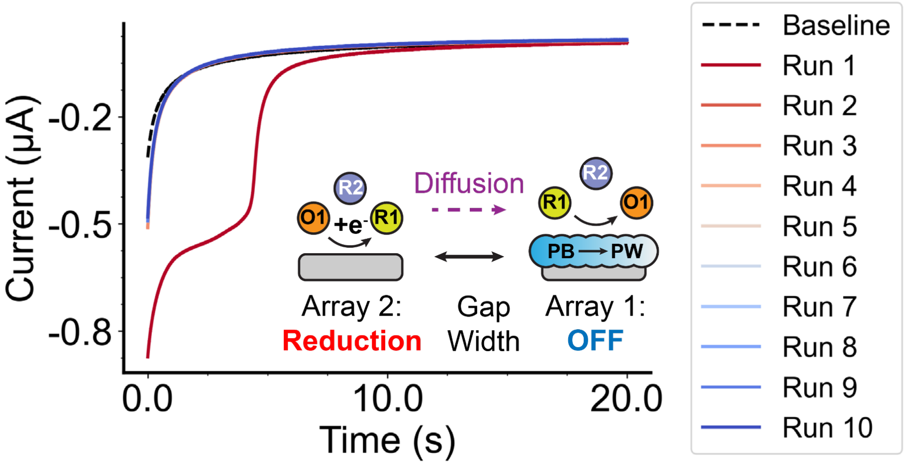
- K^+ and Na^+ showed stable redox targeting and good capacity in PB
- Li^+ showed significant capacity decay with more CV cycles
- Cs^+ shifted PB/PW's standard reduction potential, as indicated by the bipolar current increase from Fc/Fc^+TMA
  - The appeared high capacity is also due to the bipolar current increase

ART Chronoamperometries (CAs): Workflow

Baseline CA (reduction)

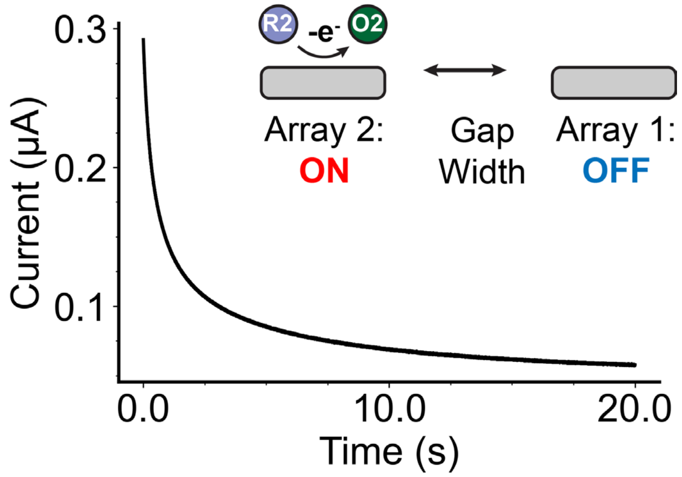


Charging DMRT CAs (reduction)

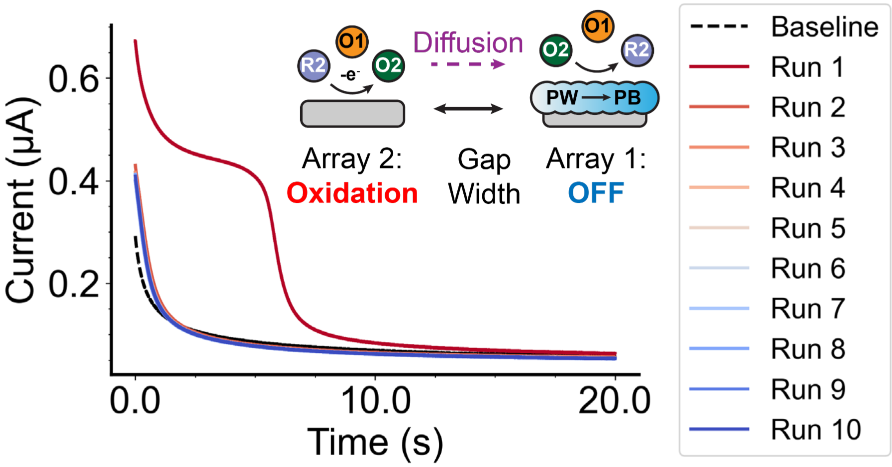


- 10 consecutive CAs to fully reduce PB to PW
- Argon bubbling in-between to reset the initial condition

Baseline CA (oxidation)



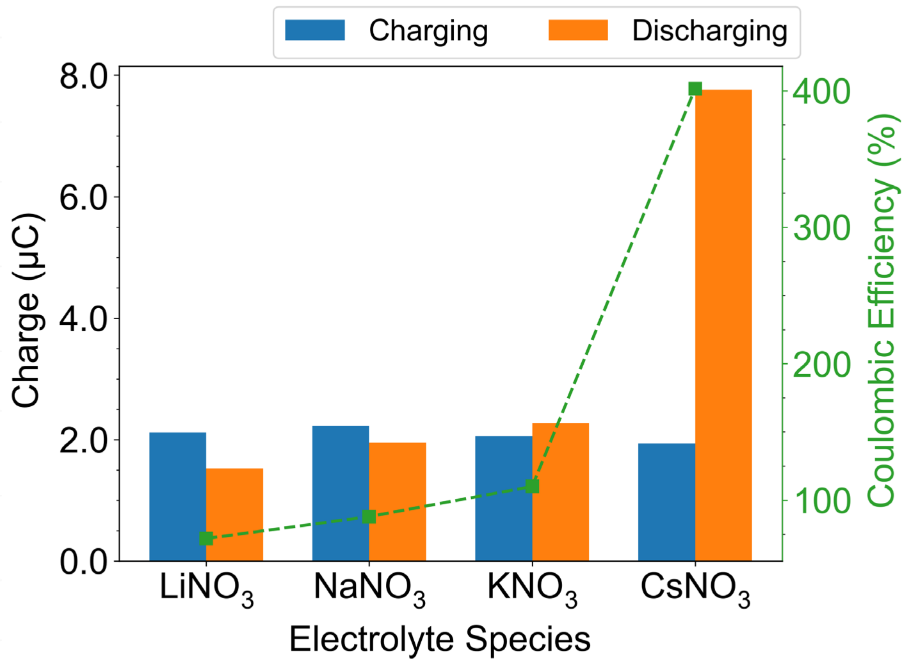
Discharging DMRT CAs (oxidation)



- 10 consecutive CAs to oxidize PW back to PB

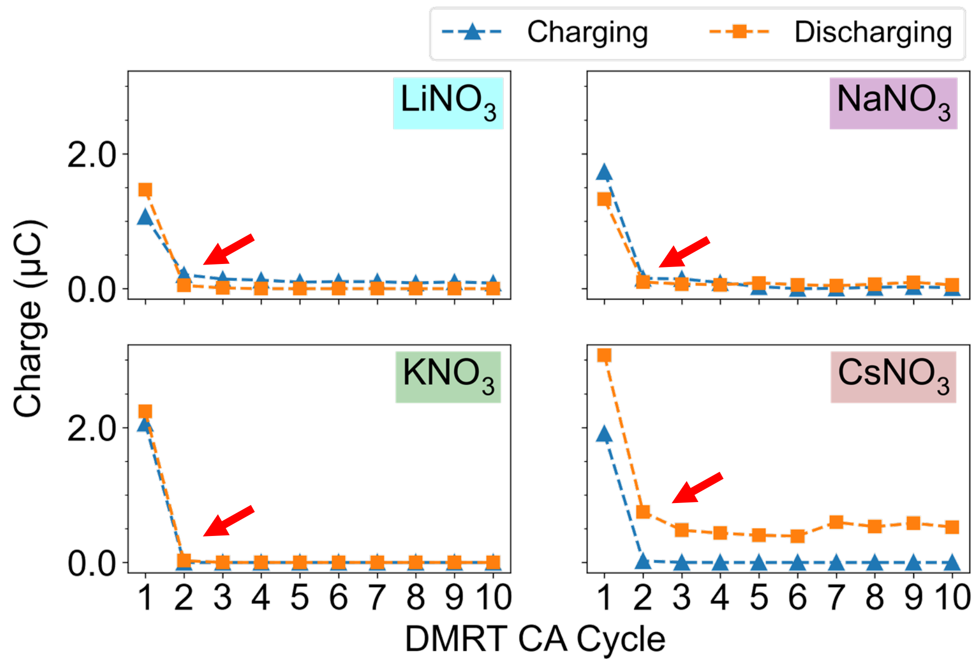
ART Chronoamperometries (CAs): Kinetics Information?

Capacity and efficiency



- **Na⁺ and K⁺ showed great capacity**
- Li⁺ showed capacity retention, while Cs⁺ showed larger capacity due to bipolar effect

Capacity accessed by each individual CA



- K⁺ and Na⁺ capacity converged to background the fastest
- **We are open to collaborations & suggestions** for modeling the redox targeting reaction to get kinetics information!

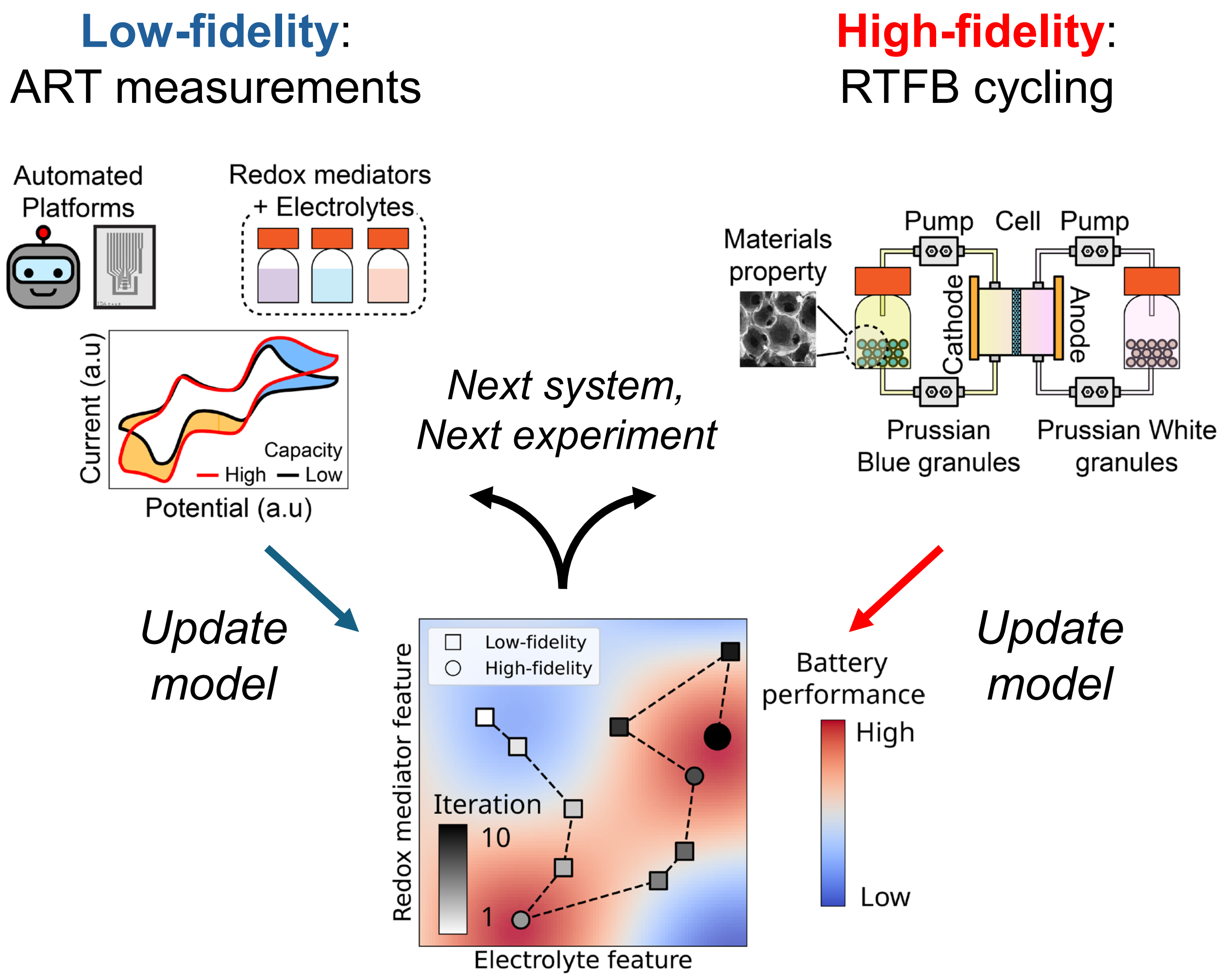
Summary and Outlook

Conclusion:

- ART is a high-throughput, modular, and versatile that sheds light on complicated effects such as charging mechanism and electrolyte composition, which are crucial to the design and optimization of RTFB systems

Outlook:

- Couple ART with finite-element simulations and other modeling techniques to obtain kinetics information of redox targeting processes
- Explore transferability of top-performing candidates suggested by ART measurements to device-level, with AI/ML algorithm (for instance, Bayesian optimization) to enable close-loop studies of redox targeting processes



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- Dr. Raghuram Gaddam, Dr. Abhiroop Mishra
- Gavin Hazen and **Seth Putnam**
- All the 4th years
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* Coauthors of the published portion of this work are highlighted.

THANK YOU! Questions?

Check out our group's other presentations and posters at ECS!

Check out our group's website!

Connect with me on LinkedIn!

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